

EXPERIMENT 11

Diabetes Prediction using Ensemble Learning

1. Problem Definition

The objective of this experiment is to predict whether a person has diabetes based on medical attributes such as:

- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Age

This is a **binary classification problem**:

- **0 → No Diabetes**
- **1 → Diabetes**

Additionally, multiple models are compared:

- Logistic Regression (baseline)
- Decision Tree
- Bagging Classifier
- Random Forest

2. Data Understanding

Dataset Characteristics

- Samples: **768**
- Features: **8**
- Target: Outcome

Class Distribution

- No Diabetes : **500**
- Diabetes : **268**

Feature Correlation

Top predictors:

- Glucose : **0.466**
- BMI : **0.292**
- Age : **0.238**

Interpretation:

- Glucose is the **dominant factor**
- Strong medical relevance

3. Data Preprocessing

Handling Invalid Values

- Zero values replaced with median:
 - Glucose : 117
 - BMI : 32.3
 - Insulin : 125

Outlier Treatment (IQR Method)

- Insulin : **346 outliers**
- SkinThickness : **87**
- BloodPressure : **14**

4. Train-Test Split

- Train: **614 samples**

- Test: **154 samples**
- Stratified split preserves class ratio

5. Models Used

Model	Type
Logistic Regression	Linear
Decision Tree	Rule-based
Bagging	Ensemble
Random Forest	Ensemble

6. Evaluation and Prediction

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Cross-validation Accuracy

Evaluation Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	CV Accuracy
Logistic Regression	0.71	0.60	0.50	0.55	0.807	0.771
Decision Tree	0.77	0.65	0.74	0.6897	0.797	0.722
Bagging	0.75	0.66	0.57	0.6139	0.8248	0.759
Random Forest	0.73	0.64	0.52	0.5714	0.8156	0.760

Confusion Matrix Insights

Key Pattern:

- All models detect **No Diabetes better than Diabetes**
- Minority class (Diabetes) is harder to classify

Reason:

- Class imbalance

7. Model Comparison Analysis

1. Logistic Regression (Weak Baseline)

- F1-score = **0.55**
- Recall = **0.50**

Misses many diabetes cases

Linear model insufficient

2. Decision Tree (Best F1-score)

- F1-score = **0.6897 (highest)**
- Recall = **0.74 (best detection of diabetes)**

Best at identifying positive cases

BUT:

- CV accuracy = **0.72 (lowest stability)**

Overfitting risk

3. Bagging Classifier (Best Balance)

- Highest ROC-AUC = **0.8248**
- Good CV stability

More generalizable than Decision Tree

4. Random Forest

- Good CV accuracy (~0.76)
- Moderate F1-score

Balanced but not optimal

ROC Curve Analysis

- Bagging has highest AUC → best separability
- Logistic Regression surprisingly competitive
- Decision Tree slightly lower AUC
- Ensemble methods improve performance

Feature Importance

- Glucose consistently highest importance
- BMI and Age also significant

Best Model Selection (CORRECT WAY)

There is **no single “best” model** — depends on goal:

Goal	Best Model
Max recall (detect disease)	Decision Tree
Best generalization	Bagging
Balanced performance	Random Forest